

OPSI45: INTRODUCTION TO UNIX / LINUX AND THE INTERNET

WEEK 1: LESSON 2

ISSUING LINUX COMMANDS / LINUX COMMAND HELP
COMMAND LINE EDITING / ONLINE TUTORIALS / SLGS

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LESSON 2 TOPICS

Using Your Matrix Account

- Issuing Linux Commands / Arguments / Options
- Command Help / Command Line Editing
- General Linux Commands

Getting Practice Issuing Linux Commands

- Performing Weekly Tutorials For Marks
- Linux Practice Questions
- Getting Help / SLGs (Student Learning Groups)

Homework

- Perform **Tutorial 1 – Investigation #2**



USING YOUR MATRIX ACCOUNT

Linux Command Structure

command **argument1** **argument2** ...

Some Linux commands can be issued by entering the Linux command line **without arguments** (e.g. **pwd**, **date**, **ls**, **cal**), but some Linux commands can be issued with arguments (e.g. **cal 2002**, **cd /bin**, **ls -la**).

An **argument** can be a file **pathname**, **text**, or an **option**.

Examples:

- The **ls** command displays a listing of just filenames in the **current** directory
- The **ls /etc** command displays a listing of filenames in the **/etc** directory (as opposed to your current directory)
- The **ls -l** command displays a **detailed** listing of filenames in the **current** directory
- The **ls -l /etc** command displays a **detailed** listing of files in the **/etc** directory



USING YOUR MATRIX ACCOUNT

Getting Help with Linux Commands

With the Linux OS containing over **2500** commands and utilities, it is good for a Linux user or Linux System Administrator (i.e. sysadmin) to learn about how to use commands “*on-the-fly*”.

The **man** command can provide information on how to use a command (i.e. **usage, arguments, options, examples**). The commands are classified into sections or “**volumes**”.

Example:

man ls

If you do not know the name of a Linux command, the **man** utility can be used with the **-k** option to help list Linux commands that match a text pattern that is contained within the help screen for a Linux command.

Example:

man -k copy



USING YOUR MATRIX ACCOUNT

Getting Help with Linux Commands / Continued...

You can use the following short-cut keys within the **man** command to help navigate throughout this utility to get help with the specific command.

Keyboard Shortcut	Purpose
ENTER	Move down one line
SPACEBAR	Move one screen down
<ctrl>	Move one screen up
/pattern	Search for Pattern
q	quit man utility

MANAGING DIRECTORIES



Instructor Demonstration

Your instructor will demonstrate how to use the **man** pages.



USING YOUR MATRIX ACCOUNT

General Linux Commands

Your instructor will demonstrate several basic Linux commands to get practice how to issue **Linux commands** and using **arguments** and **options**.

Shortcut Key(s)	Arguments / Options	Purpose
pwd		Display Current Working Directory
cd	dir-pathname	Change Directory
ls	-l, -a, -R, -d, dir-pathname	List Files of Directory
cal	month, year	Display calendar
date		Display date and time
who		List users logged into server
whoami		Display username of user logged in
clear		Clear Screen
passwd	username	Change user's password

MANAGING DIRECTORIES



Instructor Demonstration

Your instructor will demonstrate how to **issue general Linux commands**.

USING YOUR MATRIX ACCOUNT



Command Line Editing

Learning **shortcut keys** in any OS terminal will allow you to be more productive as a sysadmin. We will only focus on a few command line editing keyboard shortcut keys.

Shortcut Key(s)	Purpose
<ctrl><l>	Clear Screen
<ctrl><u>	Clear Command Line
<Up Arrow> , <Down Arrow>	Scroll Up / Down Command History
<backspace> , <ctrl><backspace> , <ctrl><h>	Delete character before the cursor
<ctrl><w>	Delete word before the cursor
<ctrl><a>	Move cursor to beginning of command line
<ctrl><e>	Move cursor to end of command line
<alt>f/<alt>b (Mac: OPTION +Right/Left-Arrow)	Move Forward/Backward one word

NOTE:

If you are using a **Graphical SSH application**, you may need to configure the application (META settings) to NOT bring up menus by mistake when you issue some of these shortcuts.

MANAGING DIRECTORIES



Instructor Demonstration

Your instructor will demonstrate how to perform **command line editing**.

GETTING PRACTICE ISSUING LINUX COMMANDS



Weekly Tutorials / Linux Practice Questions

There are **Weekly Tutorials** that are required to be completed by students for a **2%** grade for each tutorial. These tutorials are usually due by the following week by Friday @ midnight.

These tutorials are designed to provide you **guided hands-on practice** with Linux commands and operations that will help you get **troubleshooting** practice.

NOTE: Students that do NOT complete ALL parts of each weekly tutorial will NOT obtain the full 2% grade.

Linux Practice Questions are at the end of each weekly tutorial. Although these practice Linux questions are NOT for marks, they are useful for studying for quizzes and tests.

Method 1: Connecting To Your Matrix Account From Your Home Computer

You will now learn a quick and simple method of connecting to your Matrix account by opening a text-based **terminal** (in any OS) and issuing the **ssh** command.

Perform the Following Steps:

1. If you want to connect to your Matrix account from home, you **MUST** first connect to the Seneca Student VPN. Make certain that your **Seneca Student VPN** is connected.

- **NOTE:** If you haven't set this up, refer to the following link for instructions:
<https://students.senecapolytechnic.ca/spaces/186/it-services/wiki/view/1024/vpn>
- You only have to connect to your Seneca student VPN once during the day, or while you are logged into your computer.

GETTING PRACTICE ISSUING LINUX COMMANDS



Review Tutorials

Is a **Review Tutorial module** that students perform to answer questions to test their Unix/Linux Knowledge. The review tutorial is **worth a total of 10%** which is broken down into **2 general sections** worth **5%**.

As students correctly answer questions within a section, they can proceed to the next section. If the student cannot answer a question, they cannot proceed to the next question.

Students will need to successfully complete ALL sections in order to receive the 10% grade. This review tutorial module is **due at the end of the semester** (refer to the Weekly Schedule for the due date).

```
(1/1) Menu

What section you want to play next? Enter that number (1 - 3, 6 - 10):

1 - Lab navigation
2 - Introduction
3 - Vim and vimtutor
6 - File Permissions
7 - Directory Permissions
8 - g/re/p and regular expressions
9 - sed and awk
10 - Scripting

Last Update: Thu, Dec 16 at 05:55 pm EST approximately
```


GETTING PRACTICE ISSUING LINUX COMMANDS



Need Additional Help? Try the Learning Centre:

<https://library.senecacollege.ca/learningcentre>

ONE-ON-ONE TUTORING

Appointments focused on your individual needs that explain course concepts.

SUPPORTED LEARNING GROUPS (SLG)

Student-led and collaborative study sessions that review practical examples based on the course's content. Link: <https://library.senecacollege.ca/learningcentre/slg>

ENGLISH LANGUAGE SUPPORT

Offered through individual appointments or group learning sessions to focus on grammar, academic writing, conversation, and pronunciation.

STUDY SKILLS

Learn time management, exam preparation, critical thinking, note-taking, and reading.



Welcome to the Learning Centre

We are here to support you academically.
We offer services in both foundational skills and foundational course support.

Online Services & Resources

As a Seneca student, you are able to access our group study sessions, workshops, and peer tutoring at no extra cost to you.

We also have a large number of other online resources available to help you do your very best.



NEED HELP WITH YOUR COURSE?



NEED HELP WITH WRITING AND ENGLISH?



Command History

Command History:

The `~/.bash_history` file stores recently executed command lines.

There are several techniques using the `~/.bash_history` file to run previously-issued commands..

Examples:

`<up>` or `<down>` move to **previous** or **next** command in Bash shell prompt

`fc -l` display last **16** commands

`history | more` display all stored commands

`!#` **re-executes** command by command number
(obtained from *history* command)

`!abc` **re-executes** last command beginning with string
`"abc"`

Command History

Instructor Demonstration

Your instructor will now demonstrate how to use and access your **command history**.



HOMEWORK

1. Get acquainted with the **OPSI 45 Lab website, notes, tutorials and resources.**
2. Perform the following investigations in **Tutorial #1**
(Due: Friday Week 2 @ midnight for a 2% grade):
 - [INVESTIGATION 2: USING THE LINUX SHELL / ONLINE ASSIGNMENTS](#)
 - [LINUX PRACTICE QUESTIONS](#) (Questions 1 – 9)